

Stock Index Prediction Based On VMD-ARIMA-CNN-LSTM

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Abstract—Given that stock data are characterized by non-smoothness and nonlinearity, stock prediction has been a hot financial issue worth studying. To address this problem, this paper considers constructing a traditional sequence prediction model superimposed with deep network residual correction to improve the performance of stock prediction. Firstly, the Kalman filter denoising technique is used to smooth the stock data, and then VMD decomposition based on minimum sorting entropy and model complexity adaptive parameterization is considered to weaken the complexity of sequence data, and then according to the sub modularity, ARIMA-CNN-LSTM residual correction model is constructed to complete the prediction. Experiments show that the accuracy and stability of the model have good reference significance.

CCS CONCEPTS : *Applied computing~Law, social and behavioral sciences~Economics*

Keywords : *Stock Prediction, Kalman filter, VMD, ARIMA-CNN-LSTM Residual Correction Model*

I. INTRODUCTION

With the continuous development of China's financial industry, stock market forecasting has become a key reality in the research of this industry, and more and more scholars have invested in the research of stock forecasting in order to realize the continuous improvement of the accuracy of stock forecasting in the financial market. Meanwhile, the methods of stock forecasting are still improving. Whether Zhao Kangyin et al. (2020) establish an ARIMA model to predict the closing price of TCL Group on September 17, 2019, by using the group's historical stock data in 2019 as a training set; or Chen Dengjian et al. (2022) who use a combination model combining the autoregressive differential moving average model (ARIMA) with the rolling residual model of the support vector machine (SVR) for stock data forecasting; or Xie et al. (2023) proposed the EMD-SSA-LSTM-SVR combination model, which has higher accuracy than other existing models based on EMD and SSA.

Now, we consider taking the Dow Jones Industrial Average stock index as an example, firstly using the Kalman filter denoising technique to smooth the stock data, and then considering VMD decomposition based on the minimum sorting entropy and model complexity adaptive tuning parameter to weaken the complexity of the sequence data, and then based on the decomposition modality, constructing the ARIMA-CNN-LSTM residual correction model, completing the stock short-term high-precision prediction, to assist stock investment decisions.

II. DATA ANALYSIS

A. Data Description

Stock data: Dow Jones Industrial Average (DJIA) The detailed information can be seen in the table1 below. The meaning of the indicators in the table1 can be seen in the table2.

B. Data Processing

1. Missing Statistical Variables and Filling

In the process of processing the data, we perform data statute and cleaning and detect whether there are missing values, duplicate values, or outliers. The final processing results are as follows:

-No missing values

-No duplicate values

-contains outliers (i.e., values that deviate significantly from the neighboring historical data), but considering that the stock itself is affected by multiple factors and the market fluctuates greatly, it retains the outliers for feature extraction and analysis

2. Adjustment of time ordering of datasets

During data cleaning and statute, it was noted that the stock data showed an inverted order (starting from date 31-12-2015 to date 11-12-2015, from Table1), so it was considered first to perform the time-indexed sequential ordering to facilitate the subsequent proposed time series analysis and forecasting.

TABLE I. DOW JONES INDUSTRIAL AVERAGE (DJIA) STOCK

Date	Open	High	Low
31-12-2015	17590.66016	17590.66016	17421.16016
30-12-2015	17711.93945	17714.13086	17588.86914
29-12-2015	17547.36914	17750.01953	17547.36914
28-12-2015	17535.66016	17536.90039	17437.33984
24-12-2015	17593.25977	17606.33984	17543.94922
23-12-2015	17427.63086	17607.91992	17427.63086
22-12-2015	17253.55078	17451.10938	17242.85938
21-12-2015	17154.93945	17272.35938	17116.73047
18-12-2015	17495.03906	17496.58008	17124.31055
...	...	...	...
Date	Close	Volume	Adj Close
31-12-2015	17425.0293	93690000	17425.0293
30-12-2015	17603.86914	59760000	17603.86914
29-12-2015	17720.98047	69860000	17720.98047
28-12-2015	17528.26953	59770000	17528.26953
24-12-2015	17552.16992	40350000	17552.16992
23-12-2015	17602.60938	92820000	17602.60938
22-12-2015	17417.26953	91570000	17417.26953
21-12-2015	17251.61914	114910000	17251.61914
18-12-2015	17128.55078	344560000	17128.55078
...	...	...	...

TABLE II. MEANING OF DATA SET INDICATORS

Column	Meaning
Date	Dates
Open	Opening price of a stock
High	Highest stock price on that particular date
Low	The lowest stock price for that particular date
Close	Close price of a stock
Volume	Number of registered transactions
Adj Close	Adjusted closing price

III. STOCKS TIME SERIES ANALYSIS PREPARE YOUR PAPER BEFORE STYLIN

This part considers smoothness, white Noise, autocorrelation, partial autocorrelation, and preliminary forecasting.

Taking the "close" closing price of the Dow Jones Industrial Average (DJIA) as an example, the stock data are subjected to the ADF test for stationarity, Ljung-Box test for white noise, autocorrelation, and partial autocorrelation analyses, as well as recursive plotting.

A. Smothness ADF test, white noise Ljung-Box test, and autocorrelation and partial autocorrelation analysis

The ADF test, or unit root test, is a test for the presence of a unit root in a series since the presence of a unit root is a non-stationary time series.

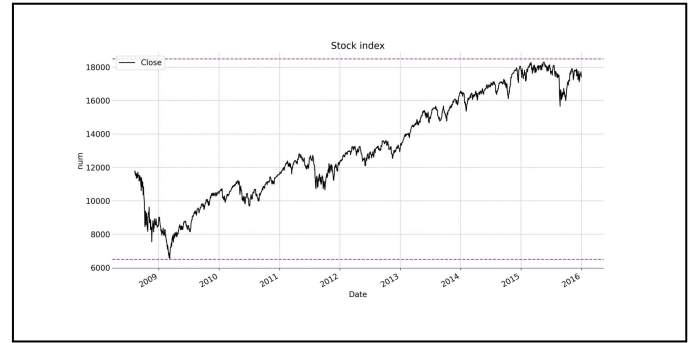


Figure 1. Correlation Diagram

TABLE III. RAW STOCK CLOSE DATA

Test and analysis	statistical value	P-value
ADF test (before first-order difference)	-0.252261	0.931999
ADF test (After first-order difference)	-10.171968	7.051090e-18
Ljung-Box test	...	lb_pvalue all $\rightarrow$ 0.0

The Ljung-Box test, i.e., LB test, and randomness test, is used to test whether the autocorrelation of the series in the m th order lag range is significant or whether the series is white noise. If it is white noise data, i.e., randomly perturbed data, the data has no value to extract, which means there is no need to continue the analysis.

Autocorrelation is simply the correlation between the series and its lag. If the series is significantly autocorrelated, the mean and the previous value of the series (the lag) may be useful in predicting the current value.

The partial autocorrelation also conveys similar information, but it conveys a pure correlation between the sequence and its lags, excluding the contribution of other intermediate lags to the correlation.

This example creates a 2D plan view showing delay values along the x-axis and correlations (between -1 and 1) on the y-axis. The confidence intervals are drawn as cones. By default, the confidence interval is set to 95%.

The results are presented in Figure 1 and Table3:

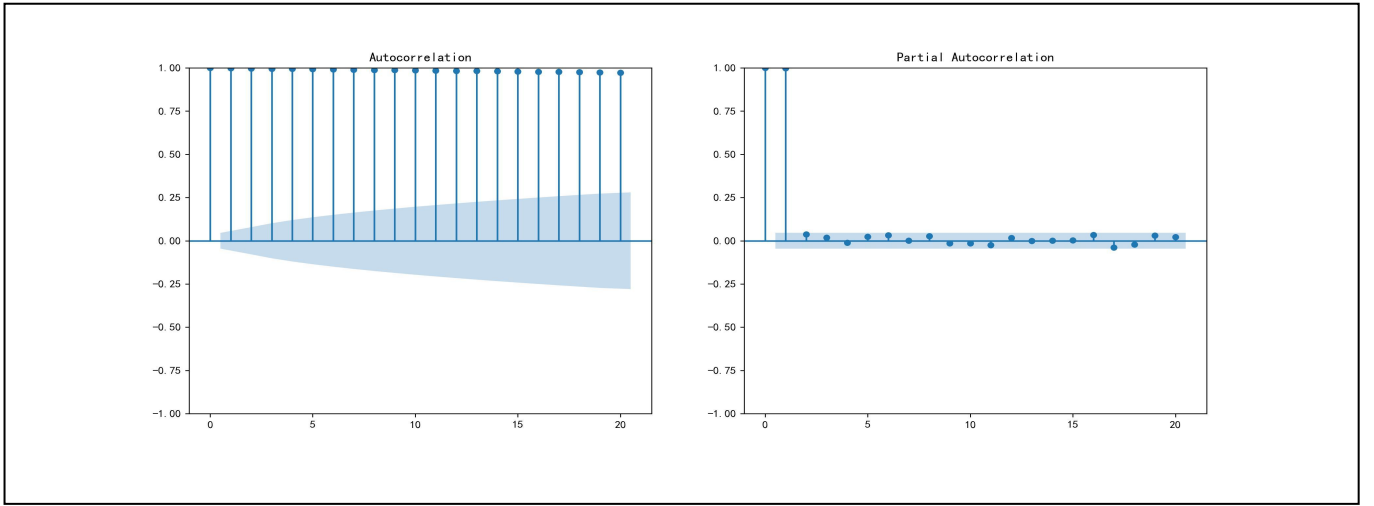


Figure 2. Visualization of raw stock autocorrelation and partial autocorrelation

From the above results, we can see that the original stock data is not a smooth series, so first-order difference smoothing is considered; secondly, through the Ljung-Box test, we know that the stock data is not white noise data, which is meaningful to explore.

From the above Figure2, it can be seen that the data possesses a certain autocorrelation strength, which can be considered as a preliminary exploration of the prediction using the autocorrelation prediction model.

B. Recursive graphs and post-differential sequence visualization

Recurrence Plots (RP) is a method for visualizing and analyzing time series or dynamic systems. It transforms a time series into a graphical representation to analyze repeating patterns and structures in the time series. Recurrence Plots are more useful on time series data, especially when there are periodicities, repeating events, or correlation structures in the time series data.

The basic principle of Recurrence Plots is to measure the similarity between points in a time series. If the distance between two time points is less than a given threshold, a point is plotted in the Recurrence Plot indicating that there is a repetition between the two time points. These points form a kind of image on a two-dimensional plane.

Choosing an appropriate similarity threshold is a key step in recursive graph analysis. Smaller thresholds usually result in more repetitive patterns, while larger thresholds result in fewer repetitive patterns. The choice of thresholds usually needs to be adjusted according to the characteristics of the data and the goals of the analysis. In this paper, we adjusted the threshold to get a final result of 10.

From the recurrence plot(Figure 3), it appears that the stock series is not a chaotic time series or a random wandering series, so there is no need to use granularity reduction methods for series prediction.

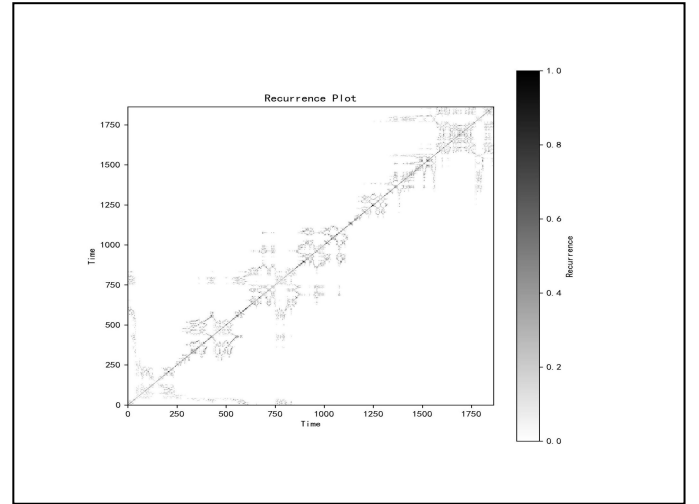


Figure 3. Recurrence Plots

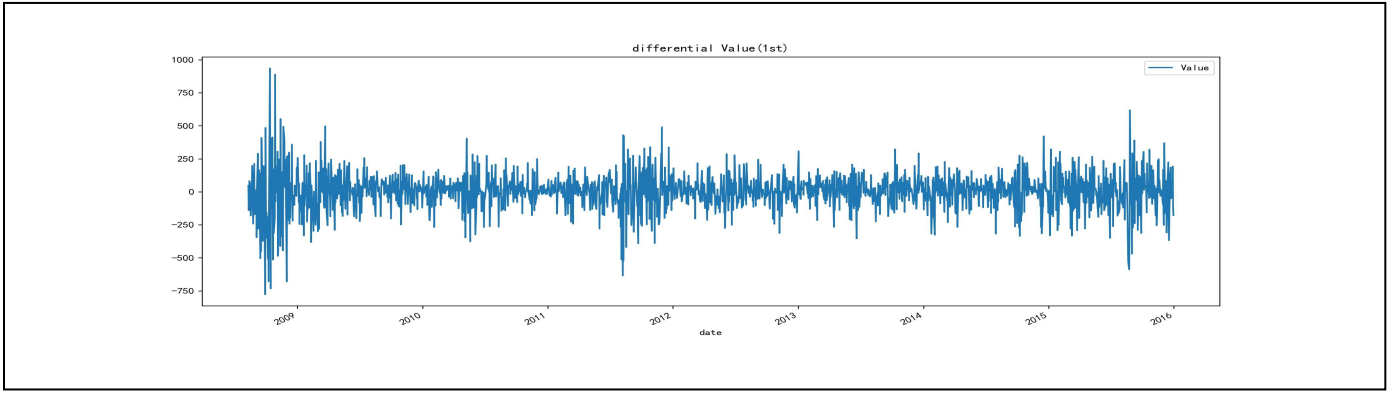


Figure 4. Differenced stock time series (smooth)

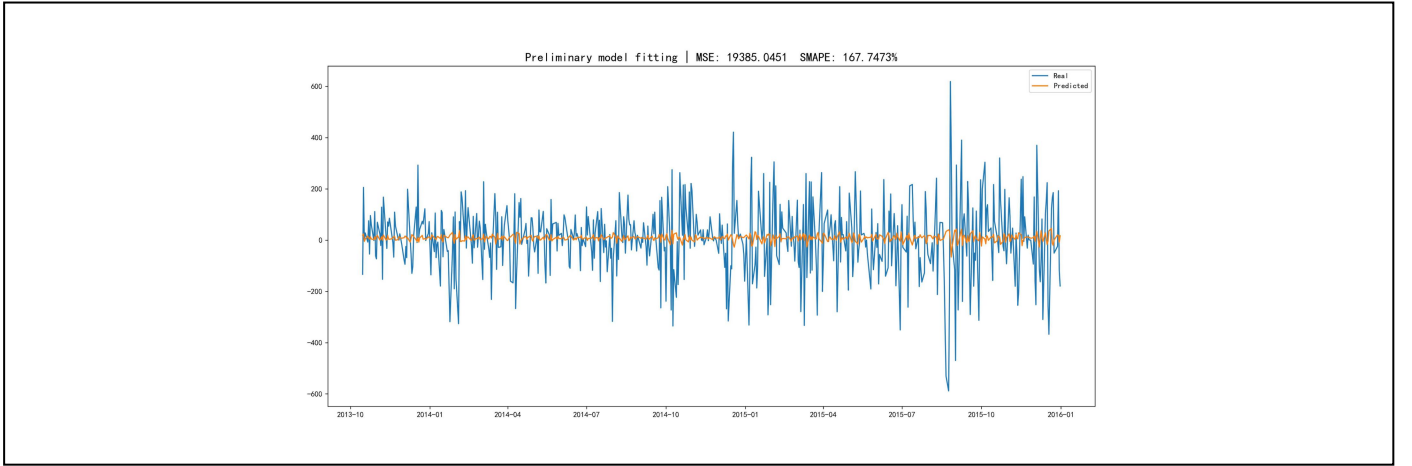


Figure 5. Fitting of the time series

C. Preliminary Prediction Exploration (SVM fitting based on the sliding window)

Considering the large autocorrelation of the data and the smoothness of the time series after differencing, we first try to fit the time series directly using sliding window based SVM. The results are shown in Figure 5:

-MSE: 19385.0451

-MAPE: inf (0 value exists)

-SMAPE: 167.7473

From the results of the above figures and the above fitting effect evaluation indexes, the model fitting effect is extremely poor, considering that it may be the higher complexity of the time series that causes the base model can not learn better, we need to improve the model to correct.

IV. FORECASTING MODEL CORRECTION ENHANCEMENTS

A. Smoothing technique based on Kalman filter denoising

Kalman filter (Kalman filter) is an efficient autoregressive filter, which is a powerful and versatile tool for estimating the state of a dynamic system in combinatorial information where there are many uncertain situations. The Kalman filter ~ raw data fitting visualization can be seen in Figure 6.

B. VMD Decomposition Based on Minimum Sorting Entropy and Model Complexity Adaptive Tuning Parameters

VMD (Variational Modal Decomposition) is an adaptive, fully non-recursive approach to modal variability and signal processing. The technique has the advantage of determining the number of modal decompositions, and its adaptivity is manifested in the fact that the number of modal decompositions of a given sequence is determined according to the actual situation, and the subsequent searching and solving process can

adaptively match the optimal center frequency and finite bandwidth of each modality and realize the effective separation of the intrinsic modal components (IMFs), the division of the signal into frequency domains, and then obtain the effective decomposition components of the given signal, and finally obtain the optimal solution of the variational decomposition problem. The optimal solution of the variational

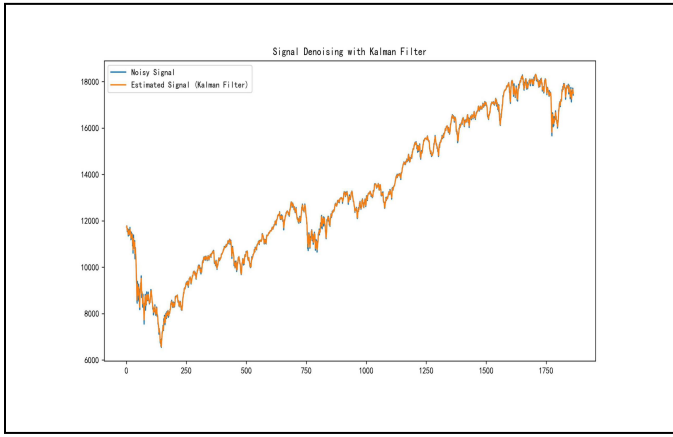


Figure 6. Kalman filter ~ raw data fitting visualization

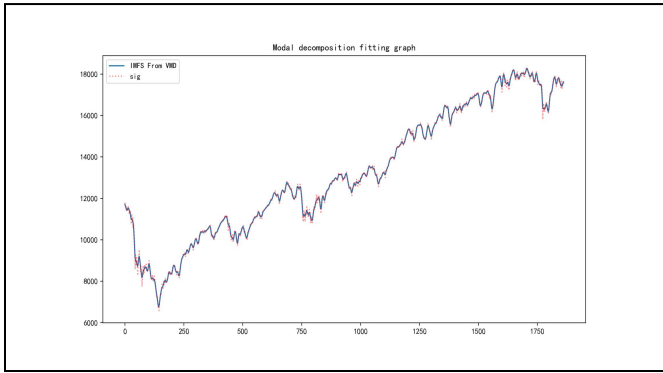


Figure 7. Model decomposition fitting graph

problem is finally obtained. It overcomes the problems of endpoint effect and modal component overlapping in the EMD method and has a more solid mathematical theory foundation, which can reduce the non-stationarity of time series with high complexity, and nonlinearity, and decompose the time series to obtain a relatively smooth subsequence with multiple different frequency scales, which is suitable for non-stationary series.^[1]

Assuming that the original signal f is decomposed into k components and the decomposition sequence is guaranteed to be a finite bandwidth modal component with a center frequency while the sum of the estimated bandwidths of the modes is minimized, and the constraints are that the sum of all the modes is equal to the original signal, the VMD constrained variational model is as follows:

$$\min_{\{u_k, \omega_k\}} \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \quad (1)$$

$$\text{s.t.} \quad \sum_k u_k = f$$

where $u_k = \{u_1, u_2, \dots, u_k\}$ is the function of each modality, and $\omega_k = \{\omega_1, \omega_2, \dots, \omega_k\}$ is the center probability of each mode.

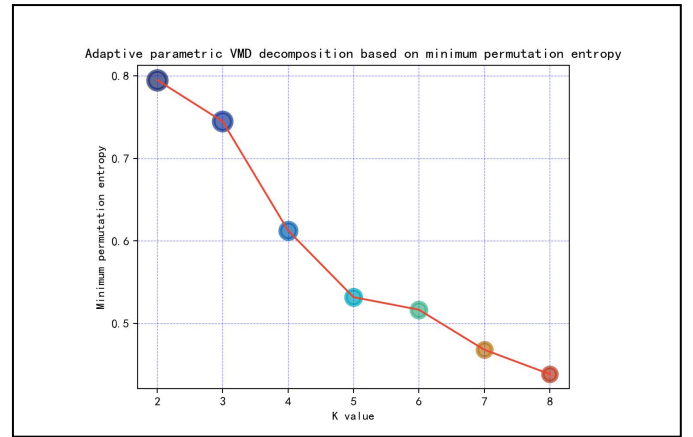


Figure 8. Visualization of adaptive parametric VMD decomposition based on permutation entropy

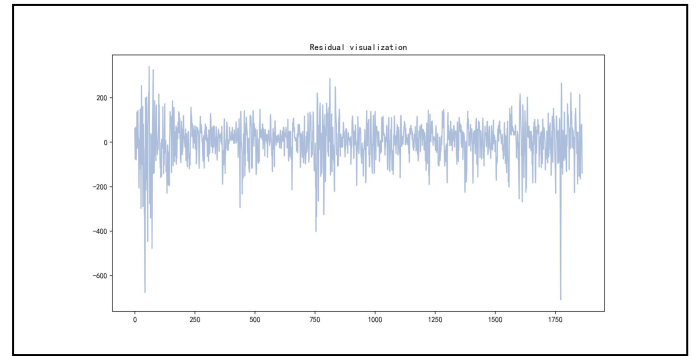


Figure 9. Visualization of residual

Since the VMD model contains hyperparameters, the choice of hyperparameters will have an impact on the effectiveness of time series decomposition. Among them, the K value represents the number of modes to be decomposed, if the K value is too large, too many modes are decomposed, making the algorithm complexity too high.

Consider the minimum ordering entropy (the value of the ordering entropy can effectively reflect the complexity of the time series, and the ordering entropy can better reflect the regularity of the time series after normalization. The smaller the entropy is, the more regular the time series is; on the contrary, the more random the time series is):

$$H = \frac{\left(- \sum_{i=1}^n P_i * \log P_i \right)}{\log(m!)} \quad (2)$$

In this regard, we use the minimum sorting entropy and combine it with the algorithm complexity to construct a metric, weighting the minimum sorting entropy with α and the algorithm complexity with $1-\alpha$.

The adaptation indicators are constructed as follows:

$$F = \alpha * H + (1 - \alpha) * \Omega(K) \quad (3)$$

Where $\Omega(K)$ denotes the complexity of the predictive model under decomposition.

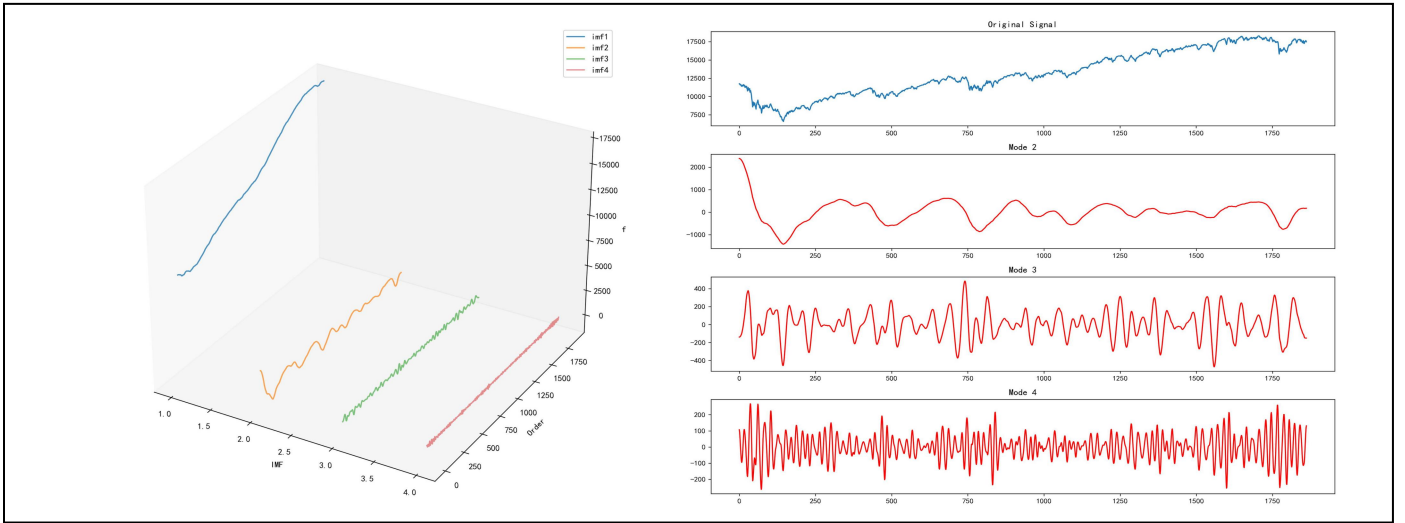


Figure 10. 3D Visualization of all modes and raw data and each imf

Considering the complexity of the algorithm and the interpretability of the model and the result from Figure 7, we finally choose $K=4$ as the hyperparameter of VMD.

View VMD decomposition effects and visualize decomposition residuals from Figure 8.

And in Figure 9, except for two very extreme points (-600) residuals, the rest of the residuals are locally stabilized at $[-400, 200]$, and the Kalman filtered data are basically above 8000, i.e., most of the residuals account for less than 5%. The effect is more significant.

Therefore, with $K=4$ as the VMD decomposition parameter, the decomposition modes imf are as follows (In the decomposition, imf1 shows an upward trend, and the rest of the modes are considered for complex periodic decomposition. Hence the value of imf1 is larger. (In Figure 10, consider the left figure for 3D visualization of all modes and the right figure for raw data and imf2, imf3, imf4).

C. ARIMA-CNN-LSTM Residual Correction Modeling

From the VMD decomposition results above we find that the modes imf1, imf2, and imf3 are relatively simple, but imf4 and the residuals are more complex both in terms of periodicity and noise, and therefore a two-layer model fusion is considered.

For modes imf1, imf2, and imf3, an ARIMA model predictive fit based on BIC (Bayesian Information Criterion) adaptive tuning of parameters is considered. Parameters display in Table4 and the fitting results are visualized in Figure 11.

For imf4 and residuals, consider the Fused CNN-LSTM model based on parameter tuning of the SMA mucilage algorithm for prediction correction.

1) ARIMA model based on Bayesian Information Criterion adaptive tuning parameters

Bayesian Information Criterion (BIC) is a criterion based on Bayesian decision theory. Bayesian Decision Theory is an important part of the subjective Bayesian school of induction

theory, which is to estimate the partially unknown state with subjective probability under incomplete intelligence, and then correct the probability of occurrence with the Bayesian formula, and then finally make the optimal decision by using the expected value and the corrected probability.

The general form of BIC is: $BIC = k \ln(n) - 2 \ln(L)$, where k is the number of parameters, n is the number of observations, and L is the likelihood function. Similar to AIC, BIC uses the likelihood function and the number of parameters to measure the goodness of fit of the model. However, BIC adds the number of data, n , to the likelihood function, which gives BIC an advantage when dealing with large data sets.

The full name of ARIMA is the Integrated Autoregressive Flat Moving Average Model and it is one of the most widely used methods for forecasting univariate time series data, which is implemented by requiring only endogenous variables without resorting to other exogenous variables for forecasting. [2] It should be noted that to forecast time series using the ARIMA model, the data must be smooth, i.e., pass the ADF smoothness test above.

2) Residual Correction of Fused CNN-LSTM Model Based on Parameter Tuning of SMA Mucilage Algorithm

CNN is a feed-forward neural network that consists of an input layer, a convolutional layer, a pooling layer, a fully connected layer, and an output layer.

LSTM network is an improvement of recurrent neural network (RNN), which effectively solves the gradient disappearance and gradient explosion of RNN by adding a gate structure. The cellular memory unit added in the LSTM network makes the LSTM network good memory ability, which is widely used in time series prediction. [3]

ARIMA Parameters	P	d	q
Imf 1	4	1	1
Imf 2	3	1	1
Imf 3	4	1	1

TABLE IV. ARIMA PARAMETERS

CNN-LSTM Network Construction.

The structure of the multi-feature sequence classification CNN-LSTM model is shown in Figure 12. It mainly consists of the signal input layer, CNN convolutional layer, pooling layer, LSTM layer, and output layer. [4]The CNN-LSTM-based multi-input sequence prediction task process is as follows:

- 1) Normalize the input signal into a CNN convolutional layer and adaptively extract features using a wide convolutional kernel;
- 2) The extracted features undergo a pooling operation in the maximum pooling layer to reduce the data dimensionality and retain the main feature information.
- 3) The downscaled feature data is then used as the feature input to the LSTM layer to train the neural network and automatically learn the sequence features.
- 4) The Adam algorithm is utilized to back-propagate the training error and gradually update the model parameters layer by layer.
- 5) Using the Softmax activation function, the signal features are predicted to predict multi-feature input sequences.

Then, the number of convolutional layers as well as the number of pooling layers of the CNN-LSTM network is considered to be adjusted using the SMA slimeball algorithm, and the number of layers within the hidden layer of the LSTM is adjusted, and the slimeball process is shown in Figure 13

Finally, taking the fourth modal imf4 as an example, the visualization model effect is shown in Figure 14:

In the Figure 14, the upper left figure shows the decreasing loss function plot of the CNN-LSTM network, iterating from 0.02 to 0.00, the model is well-trained; the upper right figure shows, constructing the error scatter plot with the real-value (test-set) data as the X-axis, and the model-predicted data as the Y-axis, and the goodness-of-fit is 0.991, P-value $\rightarrow$ 0.00, and the fitting is excellent.

Moreover, the model fully captures the characteristics of the time series, and there is no overfitting, and the post-future forecast 30 units also retain the characteristics of the historical data, and the model performs better.

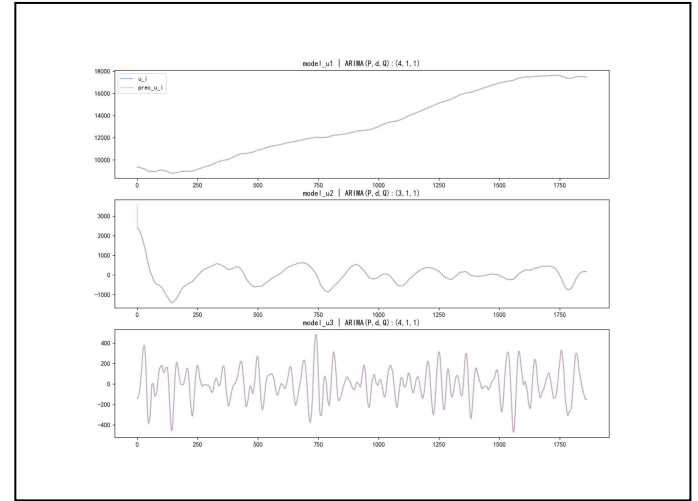


Figure 11. ARIMA for each module

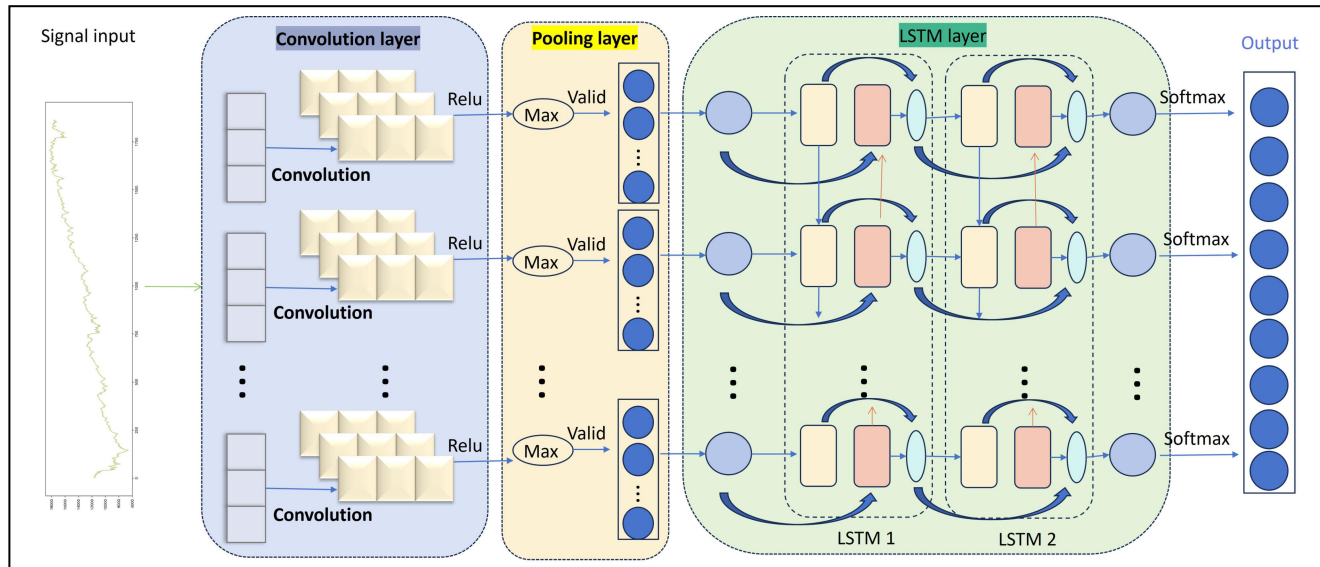


Figure 12. Visualization of CNN-LSTM network construction

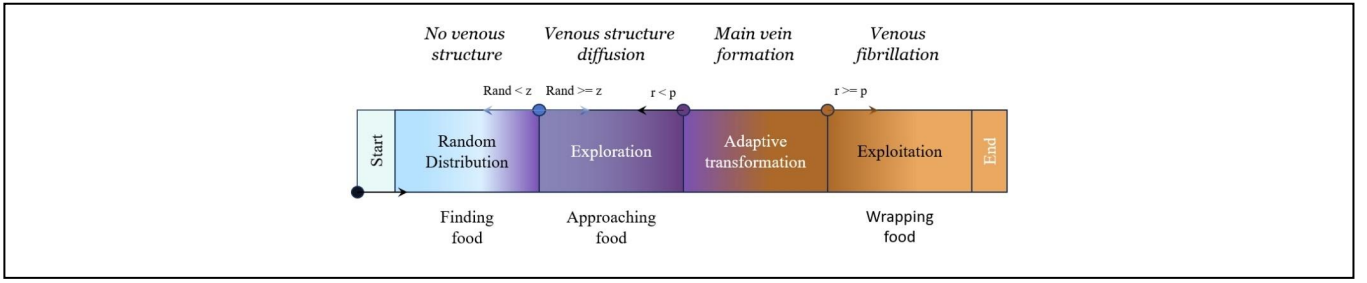


Figure 13. Visualization of the slimeball process

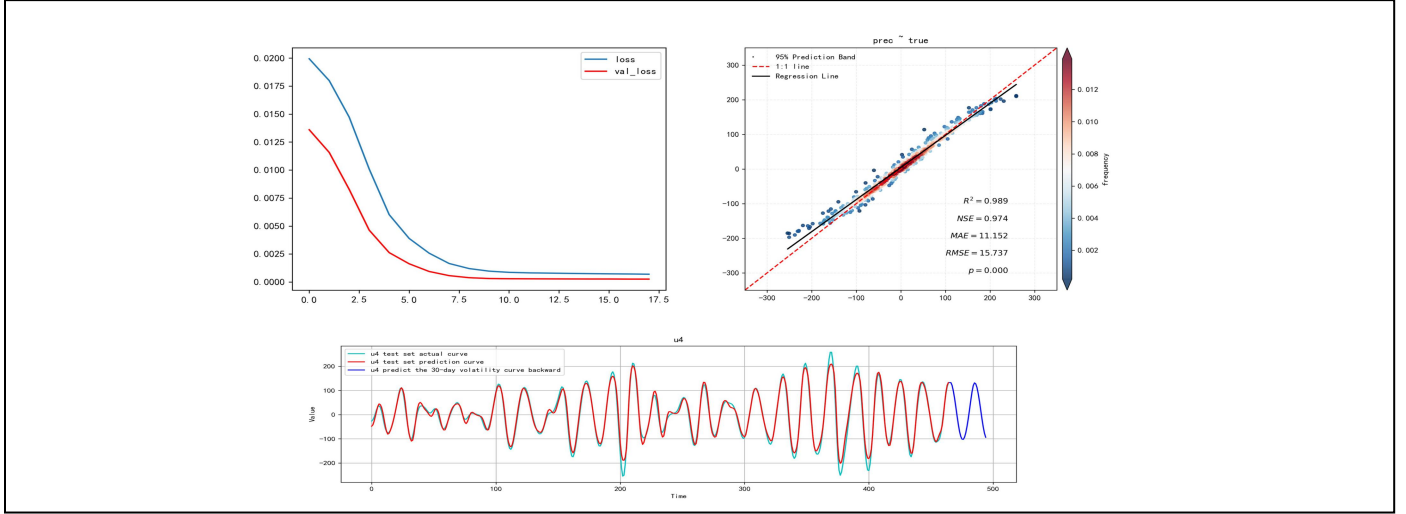


Figure 14. Visualization of model effect

Eventually, after integrating the individual modal sequences by decomposing the modal integration and reducing the individual modal sequences imf to the original stock time series, the model fit is at 0.89, and from the local modal prediction, the model grasps the trend and the complex cyclicality of the stock series and does not suffer from overfitting, which is good enough for autocorrelation sequence prediction problems.

V. FURTHER THOUGHTS ON MODELING

When comparing the Dow Jones Industrial Average (DJIA) future forecast data with the model-fit predictions, we find that the model predictions are very good in the first segment of the future forecast data, which captures the trend and extent of the stock's increase. However, at the beginning of a certain period, the model we constructed fails to predict, and the stock data shows "unnaturally" large fluctuations that are not reflected in the historical data, and therefore the model does not reflect such fluctuations.

By analyzing the "Press Kit" data, it was noted that the newsfeed then contained news that reflected unfavorable information about the company, which coincided with an "unnatural" downward trend in the stock.

Taking the above considerations into account, we can consider extracting the feature word vectors through "natural language processing NLP technology", constructing the stock environment factor, further modifying the model, introducing the self-attention mechanism, constructing the Transformer model, and further improving the reasonableness and accuracy of the stock prediction model.^[5] This will be our next research direction

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